Stefano Lacovara

Curriculum Vitae

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Professional Experience

June 2025 – present
Research Fellowship, *Università degli Studi di Padova*, Padova, Italy
Nonlinear Model Predictive Control (NMPC) for high-performance virtual car drivers
The project develops a Nonlinear Model Predictive Control for high-performance virtual car drivers, addressing complex vehicle dynamics at the limits of stability. The controller leverages a full nonlinear vehicle model with tire dynamics and lateral-longitudinal coupling, solved through real-time optimization under actuator and safety constraints, achieving near-optimal racing trajectories and robust stability.
Skills: NMPC, vehicle dynamics, CasADi, MATLAB/Simulink, C, VI-Grade suite

Education

2023–2025
M.Sc. in Control Systems Engineering, *Università degli Studi di Padova*, Padova, Italy
Grade: 110/110
Thesis: *Kinodynamic RRT* for path planning of a small fixed-wing UAV*

2018–2022
B.Sc. in Ingegneria Elettronica e Informatica, *Università degli Studi di Ferrara*, Ferrara, Italy
Grade: 96/110
Thesis: *Analisi di Casi di Studio di BCI Impiegati per il Controllo*

International Experience

April 2024 – October 2024
Master Thesis Internship, *Agile Robotics and Perception Lab, New York University*, New York, USA
Developed an enhanced Kinodynamic RRT algorithm for online path planning of small fixed-wing UAVs in obstacle-rich environments. The approach incorporated both kinematic and dynamic constraints, producing collision-free and dynamically feasible trajectories. Validated the method with real-world UAV flight tests, demonstrating robustness and applicability to surveillance, monitoring, and disaster response.
Skills: Motion planning, ROS, UAV dynamics, optimization, MATLAB, C++

Projects

November 2024 – June 2025
Bosch Future Mobility Challenge, *Bosch Engineering Center Cluj*, Cluj-Napoca, Romania
Participated in an international student competition on autonomous driving and connectivity for 1/10 scale vehicles. Designed algorithms for autonomous navigation in smart cities, obstacle avoidance, and V2X communication, and collaborated in a multidisciplinary team to integrate perception, planning, and control modules.
Skills: Autonomous navigation, ROS, sensor fusion, V2X communication, C++, Python

Languages

Italian — Native speaker

English — B2 (working proficiency; certified B2 Listening and Reading, full B2 self-assessed)

Technical Skills

Programming & Tools	C/C++, MATLAB/Simulink, Python, Bash, Git, Docker, CMake, TeX
Simulation & Modeling	CasADi, ACADOS, VI-Grade, Gazebo, RViz
Embedded Systems	ROS, PX4 Autopilot, STM32, Real-time Operative Systems concepts, sensor integration
Operating Systems	Linux (Ubuntu), macOS, Windows

Date: October 12, 2025

Signature: